Shubham Chaudhary

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Education

- **Cornell University** — Ithaca, NY
  PhD in Computer Science — *January 2020 - Present*
- **Indian Institute of Technology (BHU) Varanasi** — Varanasi, India
  B. Tech. in Computer Science and Engineering - CGPA: 9.75/10.00 - Department Rank 1 — *July 2014 - May 2018*

Research Experience

- **Microsoft Research India** — Bangalore, India
  Research Fellow — *July 2018 - December 2020*
  - Advised by Dr. Muthian Sivathanu and Dr. Ramchandran Ramjee.
  - **Gandiva**$_{fair}$ is a set of Proportionally Fair Scheduling policies for Gandiva based on the concept of using job migration to balance the ticket-adjusted load across servers and then performing intraserver proportional share to achieve cluster-level proportional share. It transparently handles GPU heterogeneity using profiling and a novel GPU trading strategy to improve cluster efficiency while preserving fairness. I was involved in the design of the scheduling algorithm and policies and their implementation.
  - **Gandiva** is a Cluster Scheduler/Resource Manager for Deep Learning Training jobs utilizing CPU Scheduling primitives like time-slicing, migration, and profiling to schedule GPUs efficiently. I redesigned and implemented the entire cluster manager using `Kubernetes` to manage job containers and wrote the scheduling framework in `Scala` using the `Akka Actors` library for concurrency and `gRPC` for performing RPCs.
  - **GPU Proxy** is a mechanism inspired from `CRCUDA` to enable suspend/resume, and migration of Deep Learning jobs written in `PyTorch` and `TensorFlow`. I implemented `Horovod` and `NCCL` support to enable suspend/resume and migration of distributed jobs.

Publications

- **S. Chaudhary**, R. Ramjee, M. Sivathanu, N. Kwatra, and S. Viswanatha. *Balancing Efficiency and Fairness in Heterogeneous GPU Clusters for Deep Learning.* 2020. In Proceedings of the Fifteenth European Conference on Computer Systems (EuroSys '20). `https://dl.acm.org/doi/abs/10.1145/3342195.3387555`

Internship

- **Goldman Sachs** — Bangalore, India
  Summer Analyst — *May 2017 - July 2017*
  - **DbBench** is a distributed Database Benchmarking system for `SecDB`, a proprietary object database, used to make critical decisions about the future of the database system. It can emulate the real-life workload of hundreds of clients accessing a single database endpoint and report aggregate metrics like throughput and transactions per second. I designed and implemented the entire system in `Scala` using `Akka Actors` library for concurrency and `Netty` for composable high-throughput, low-latency asynchronous socket IO. I also implemented a `YAML` based configuration system and an interpreted Turing Complete programming language to write performance tests in.

Selected Projects

- **LTDP for Viterbi Algorithm** `https://github.com/5hubh4m/ltdp-viterbi-algorithm` Applied the Linear Tropical Dynamic Programming model to the Viterbi algorithm to parallelise the otherwise linear algorithm using the theorem of Rank Convergence. Implemented it using `OpenMPI` in conjunction with `OpenMP` in `C++`. Investigated the improvement over vanilla LTDP by redistributing data to processors after each principal iteration to prevent otherwise redundant computation of converged rows.

- **Scheme Interpreter in Haskell** `https://github.com/5hubh4m/simple-scheme` Wrote an Interpreter for a subset of Lisp (the R5RS standard) in `Haskell`, a pure functional programming language. Implemented an REPL, a parser, variables, user-defined functions using Functional Programming concepts like maps, folds, composition, monads, do-notation, currying, pattern-matching, etc.

- **Relational Database and Relational Algebra Interpreter** `https://github.com/5hubh4m/simple_ra` Implemented a persistent Relational Database in `C++`. Hand-wrote a top-down Recursive Descent parser for interpreting a custom $LR-1$ grammar for Relational Algebra supporting operations like σ, ρ, Π, $\times$, $\cup$, $\bowtie$, $-$, $\div$, and aggregate operations.

Teaching

- **Introduction to Computer Programming** — IIT (BHU) Varanasi
  TA and Tutor — *July 2016 - May 2018*
  - Responsibilities included delivering instruction during practical sessions on programming fundamentals in `C`, providing support and resolving queries during tutorial sessions, and curating and grading assignments for a class of about 60 undergraduates.

- **Artificial Intelligence** — IIT (BHU) Varanasi
  TA — *Jan 2018 - May 2018*
  - Responsibilities included delivering instruction during practical sessions on fundamentals of Artificial Intelligence (Search, Logic, and Machine Learning), curating assignments, and building and managing a `git` based encrypted automatic submission system for their evaluation for a class of about 80 undergraduates.

Achievements

- **IIT (BHU) Varanasi Medal** for being Department Rank 1 in Computer Science and Engineering B. Tech Class of 2014-18.

- **SRFP 2016** Awarded the Summer Research Fellowship by the Indian Academy of Sciences, Bangalore.

- **Runner Up** at Microsoft's Code.Fun.Do 2016 24-Hrs Hackathon.

- **Winner** of Parliamentary Debate Competition at AMU Literary Festival, AMU Aligarh, 2015.

Positions of Responsibility

- **Event Coordinator and Problem Curator** for Capture the Flag event at Technex 2017, and CodeFest 2017, with the participation of over 500 students from India and abroad.

- **Jt. Secretary** of the Literary Club of IIT (BHU) Varanasi for the session 2015-16.

- **Web Development** Worked on the Website Development and Technical Committee of Kashiyatra 2016, FMC Weekend 2015, and IITBHUMUN 2015.

Skills and Areas of Interest

- **Languages** `C++`, `C`, `Scala`, `Java`, `Python`, `Haskell`, `C#`, `ECMAScript`

- **Technologies** `POSIX`, Web Development, Android Development, `Docker`, `Kubernetes`, LaTeX, `MATLAB`

- **Areas of Interest** Parallel, Distributed, and High Performance System

Relevant Coursework

CSO204 - Operating Systems	Compiler Design - CSE425
CSE202 - Artificial Intelligence	Computer Networks - CSE433
CSO324 - Operation Research	Information Security - CSE530
CSE361 - Database Management Systems	Network Security - CSE537
CSE371 - Parallel Computing	Optimisation Techniques - MA526

CS231n - Convolutional Neural Networks for Visual Recognition (Stanford University)